enveloped by bound double layered nuclear membrane called chromatin. Cell has a store convert for compartment for energy production. This is called mitochondria- the power houses of the cell. Endoplasmic reticulum or ER are the several interrelated membrane-bound compartments in eukaryotic cells.

Ribosomes are the granular structures composed of ribonucleic acid and proteins. Cytosol or cytoplasm is the ground substance that fills the interior of the cell. Plastids are found in all plant cells and in euglenoides. The ability to use light for sugar synthesis from water and carbondioxide is a special feature of plant cells. This process is termed as Photosyntheis is carried out in organelles called Chloroplasts. Vacuoles is the membrane bound space found in the Cytoplasm. According to Lowry Bronsted theory. Acid is a proton-donor and a base is a proton acceptor. The acidic or basic nature of a solution is measured by H^+ ion Concentration

$pH = -\log[H^+]$

Henderson-Hasselbalch equation $pH = pK_a + \log \frac{[\text{Conjugatebase}]}{[\text{Acid}]}$

A pH meter used to measure pH of asolution.

Body fluids are water solution of dissolved substances. Two types are intra cellular fluids and extracellular fluids. The colourless fluid present in the lymphatic system is called lymph. Blood is a fluid that circulates in a closed system of blood vessels in multi cellular and highly complex vertebrate animals.

Various buffer system of the blood are.

a. Bicarbonate buffer system, b. Phosphate buffer system, c. Protein buffer system and d. Hemoglobin buffer system

The body's acid base balance is maintained by buffer systems that operates in living system (lungs and kidneys).

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